

than the half angle of resolution.

18 B

Using Malus' law,

$$I_2 = I_1 \cos^2 10^\circ = \frac{1}{2} I_0 \cos^2 10^\circ$$

$$I = I_2 \cos^2 34^\circ$$

where θ is the angle between the axes of polarisation of the second and third filters

$$I = I_2 \cos^2 34^\circ = \frac{1}{2} I_0 \cos^2 10^\circ \cos^2 34^\circ$$

$$= \frac{1}{2} I_0 \cos^2 10^\circ \cos^2 34^\circ = 0.33 I_0$$

[Turn over

19 D The path of the particle indicates that it experiences an upward force. A negatively